A Water-Inventory Governor: Passive Hydraulic Speed Regulation Without Flyweights

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Abstract—A proposed steam-engine governor converts shaft speed into water flow and combines a sharp-edged restriction with a water-inventory column. A floating siphon provides a constant reference outflow. Pressure upstream of the restriction directly moves a spring-loaded, damped steam throttle. Flow pressure supplies proportional action, while stored water volume supplies integral action. Including the actuator's displaced water gives a third-order closed loop. For normalized load steps from 1 to 1.3 and 0.7, GNU Octave simulations predict a 4.00% maximum speed excursion, recovery to within 1% of nominal in 5.55 s, and zero steady-state speed error under the ideal model. Nonlinear orifice calculations and a physical-parameter stability sweep qualify these results. Construction, calibration, and originality limitations are explicitly identified; no prototype performance is claimed.

Index Terms—Steam engine, hydraulic governor, passive feedback, integral action, stability, load rejection.

I. Introduction

A mill's common line shaft experiences sudden process-load changes when machines engage or disengage. A throttle fixed at its original opening cannot compensate automatically: an added load reduces engine speed, and removal increases it. The objective is a fully automatic governor using plausible late-eighteenth-century components, with stable recovery and low permanent speed error. The prescribed simulations use nominal speed and process-load torque equal to one.

The proposed design uses a crank-driven displacement pump, water, copper or brass pipework, a guided copper float, leather diaphragm or bellows, iron linkage, a steel spring, and a liquid dashpot. No electronics, magnetic sensing, electric motors, elastomer seals, or modern structural materials are required. Dimensions below are engineering estimates for a small mill engine; the report does not assert documented construction of this arrangement in the eighteenth century.

II. Conceptual Design

A. Sensing, memory, and throttle action

A small pump driven from the engine delivers an average flow proportional to shaft speed. Its discharge crosses a thin restriction into an open water column. A pressure tap upstream of that restriction feeds a diaphragm connected to a pressure-balanced steam throttle. Increasing water pressure closes the throttle; a spring and adjustable counterweight bias it open. The dashpot limits actuator velocity.

The column drains through a primed siphon rigidly attached to a guided float. The float carries both siphon limbs and the discharge nozzle, preserving a fixed vertical distance between the water surface and the freely discharging outlet. The outlet moves vertically outside the column over a broad collecting trough. Water returns to the pump sump. The principle of constant discharge from a floating siphon is documented in USGS flow-measurement literature [2]; its use here as the governor's flow reference is a proposed application.

When the engine slows, pump flow decreases. Restriction pressure falls immediately, opening the throttle. In addition, outflow initially exceeds inflow, lowering the column level. The resulting hydrostatic-pressure reduction keeps opening the throttle until speed returns to the calibrated value. At restored speed, inflow again equals outflow; the new water level retains the opening needed for the new load. An unloaded engine produces the opposite sequence.

B. Relationship to existing governors and novelty limits

Watt, Porter, and Proell governors sense centrifugal displacement; spring-loaded Hartnell-type designs retain this rotating-mass principle. Here shaft displacement produces flow, and accumulated fluid volume supplies the persistent

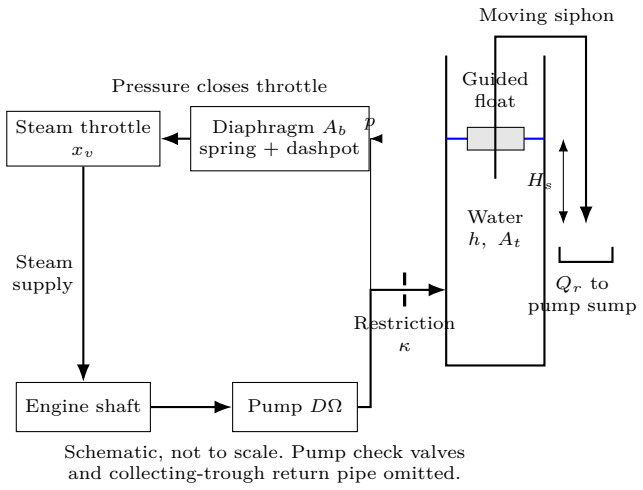


Fig. 1. Proposed physical arrangement. The moving siphon outlet must remain below the water surface and above the receiving water.

corrective state. Neither centrifugal weights nor a flyball-driven sleeve is present.

The concept shares hydraulic transmission, springs, and damping with existing hydraulic governors. For example, the mechanical section of Woodward's EGB-10PLS uses flyweights, a pilot valve, and servo/buffer pistons [3]. This proposal instead senses speed through pump flow, directly adds restriction and hydrostatic pressures, and uses a floating reference drain to retain integral memory without a flyweight-controlled pilot valve. The claimed contribution is this specific physical arrangement, not the invention of pumps, siphons, hydraulic servos, or proportional–integral control. A targeted comparison is not an exhaustive patent search; worldwide novelty is unproven.

III. Mathematical Modeling

A. Assumptions and physical equations

Let Ω be dimensional shaft speed, T_L the externally imposed process-load torque, and $x_v \in [0,1]$ the valve-opening fraction. A local averaged engine model is

$$J\dot{\Omega} = K_t x_v - B\Omega - T_L. \tag{1}$$

Steam pressure is treated as constant; valve opening gives approximately linear mean torque. Cylinder pulsations and crank-angle torque ripple are averaged. $B\Omega$ represents speed-dependent mechanical drag, separate from the specified process load T_L. Valve leakage, backlash, Coulomb friction, pump slip, and thermal changes are excluded from the nominal model and discussed later.

The pump, column, and pressure drop obey

$$Q_p = D\Omega, \qquad Q_p = q - \alpha\dot{x}_v, \tag{2}$$

$$A_t\dot{h} = q - Q_r, \qquad p = \rho g h + \kappa q|q|, \tag{3}$$

where $\alpha = A_b L$, A_b is diaphragm effective area, and L is full valve stroke referred to the diaphragm. q is the restriction flow, while Q_p is pump flow. Opening the valve reduces the pressure-chamber volume, expelling water; hence the sign in (2). Keeping this flow avoids treating the actuator as a signal-only device.

For a thin restriction with effective area A_o,

$$\kappa = \frac{\rho}{2C_d^2 A_o^2}, \qquad Q_r = C_{ds} A_s \sqrt{2gH_s} = Q_0. \tag{4}$$

These are quasi-steady Bernoulli loss models. Discharge coefficients must be calibrated. The floating siphon keeps H_s constant within its travel and free-discharge range; a fixed hole at the column bottom would not do so. A float that follows the water surface quickly and without appreciable friction is assumed.

Neglecting diaphragm inertia but including spring and viscous damping gives

$$\tau\dot{v} + v = -K_s\delta p, \quad K_s = \frac{A_b}{k_s L}, \quad \tau = \frac{c_y}{k_s}, \tag{5}$$

where $v = x_v - x_{v0}$, $\delta p = p - p_0$, and the counterweight/preload balances p_0 at the nominal opening. No constant spring force is counted twice. The quasi-static-inertia approximation requires actuator inertial dynamics appreciably faster than the modeled response.

B. Normalization and linearization

Use $\omega = \Omega/\Omega_0$, $T_L/T_{L0} = 1 + d$, $e = \omega - 1$, and $\eta = h - h_0$; time remains in seconds. At equilibrium,

$$D\Omega_0 = Q_r = Q_0, \quad K_t x_{v0} = B\Omega_0 + T_{L0}. \tag{6}$$

With $R = 2\kappa Q_0$, first-order expansion around $q = Q_0 > 0$ yields

$$j\dot{e} + be = kv - d, \tag{7}$$

$$\delta q = Q_0 e + \alpha\dot{v}, \qquad \delta p = \rho g\eta + R\delta q, \tag{8}$$

$$A_t\dot{\eta} = Q_0 e + \alpha\dot{v}, \tag{9}$$

where $j = J\Omega_0/T_{L0}$, $b = B\Omega_0/T_{L0}$, and $k = K_t/T_{L0}$. Define the stored-volume coordinate

$$z = \eta - \frac{\alpha}{A_t}v, \qquad \dot{z} = \frac{Q_0}{A_t}e. \tag{10}$$

Eliminating pressure and flow gives

$$a\dot{v} + cv = -K_P e - K_s\rho g z, \tag{11}$$

$$a = \tau + K_s R\alpha, \qquad c = 1 + \frac{K_s\rho g\alpha}{A_t}, \tag{12}$$

$$K_P = K_s R Q_0, \qquad K_I = \frac{K_s\rho g Q_0}{A_t}. \tag{13}$$

Thus proportional and integral action emerge from physical pressure and volume relations, with the actuator dynamics included in the controller. Setting $\alpha = 0$ prematurely would incorrectly give $a = \tau$ and $c = 1$.

C. Transfer functions and feedback diagram

For fixed calibrated speed reference, use $E_r = -e$ as speed error and V as controller output. With zero initial perturbations,

$$G_c(s) = \frac{V(s)}{E_r(s)} = \frac{K_P s + K_I}{s(as + c)}, \qquad G_p(s) = \frac{k}{js + b}. \tag{14}$$

The fictitious input r in Fig. 2 defines the usual reference transfer; actual setpoint changes require changing Q_r and rebiasing the mechanism. This report evaluates a fixed setpoint and load rejection.

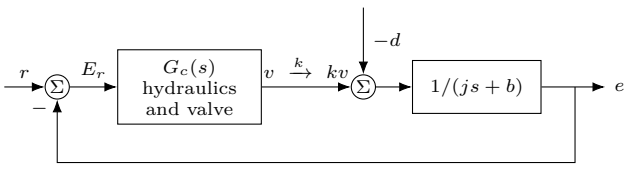


Fig. 2. Small-signal closed loop. The process-load disturbance enters as opposing torque, after the valve-to-torque gain k.

Writing

$$\mathcal{D}(s) = jas^3 + (jc + ba)s^2 + (bc + kK_P)s + kK_I, \tag{15}$$

the reference and disturbance transfers are

$$G_{CL}(s) = \frac{e}{r} = \frac{k(K_P s + K_I)}{\mathcal{D}(s)}, \tag{16}$$

$$\frac{e}{d} = -\frac{s(as + c)}{\mathcal{D}(s)}, \qquad \frac{v}{d} = \frac{K_P s + K_I}{\mathcal{D}(s)}. \tag{17}$$

The open-loop comparison fixes $v = 0$ and disconnects feedback, giving $e/d = -1/(js + b)$; it does not set engine torque to zero.

IV. Parameter Selection and Stability

A. Physically interpretable design point

Table I lists design estimates, not measured constants. For the chosen engine, $\Omega_0 = 10\,\mathrm{rad\,s^{-1}}$ is approximately 95.5 rpm; $K_t x_{v0} = 200\,\mathrm{N\,m}$ supplies 2 kW. The external process load is 100 N m and effective speed-dependent drag is another 100 N m. This substantial B assumes auxiliary belt/fan/fluid drag, not bearing friction alone. A flywheel with equivalent rim mass 40 kg at radius 1 m gives $J \simeq 40\,\mathrm{kg\,m^2}$.

TABLE I
Dimensional design estimates and their rationale

Quantity	Value	Rationale
$J;\ B$	$40\,\mathrm{kg\,m^2}$ $10\,\mathrm{N\,m\,s}$	Flywheel and effective speed drag
$\Omega_0; T_{L0}$	$10\,\mathrm{rad/s}$ $100\,\mathrm{N\,m}$	Small mill design point
$K_t; x_{v0}$	$400\,\mathrm{N\,m};\ 0.5$	Nominal torque balance
$Q_0; D$	$4\times10^{-4}\,\mathrm{m^3/s}$; $4\times10^{-5}\,\mathrm{m^3/rad}$	24 L/min; 251 mL/rev
$\rho; g$	$1000\,\mathrm{kg/m^3}$ $9.81\,\mathrm{m/s^2}$	Water near room temperature
$A_t; h_0$	$0.007848\,\mathrm{m^2}$ $0.40\,\mathrm{m}$	100 mm column bore
Δp_{o0}	1000 Pa	Moderate restriction loss
$C_d; H_s$	$0.62;\ 0.15\,\mathrm{m}$	Initial orifice/siphon estimate
$A_b; L$	$0.002\,\mathrm{m^2}$ $0.020\,\mathrm{m}$	50.5 mm diaphragm; 20 mm stroke
$k_s; c_y$	$100\,\mathrm{N/m}$ $25\,\mathrm{N\,s/m}$	Soft spring and liquid dashpot

The pressure restriction has an estimated diameter 24.10 mm; its nominal water Reynolds number is about 2.1×10^4, making a calibrated inertial-loss approximation reasonable. A 21.88 mm effective siphon outlet is an initial estimate for $C_{ds} = 0.62$; bend and pipe losses require adjusting it to obtain $Q_r = Q_0$. A thin plate and short, broad connecting pipes minimize neglected water inertance. The nominal sensed pressure is 4924 Pa, giving a diaphragm preload of 9.85 N. The pump's pressure-flow power is only 1.97 W before losses. It is supplied by the engine; the governor is not an energy source.

B. Closed-loop poles and Routh test

The estimates give $j = 4\,\mathrm{s}$, $b = 1$, $k = 4$, $K_P = 2$, $K_I = 0.5\,\mathrm{s^{-1}}$, $a = 0.45\,\mathrm{s}$, and $c = 1.05$. Consequently,

$$\mathcal{D}(s) = 1.8s^3 + 4.65s^2 + 9.05s + 2. \tag{18}$$

All coefficients are positive, and the cubic Routh condition is

$$4.65(9.05) = 42.0825 > 1.8(2) = 3.6. \tag{19}$$

The first-column entries are $1.8, 4.65, 8.2758, 2$, so every pole is in the left half-plane. More generally, stability requires

$$(jc + ba)(bc + kK_P) > ja\,kK_I. \tag{20}$$

Excessive integral gain or actuator lag can therefore destabilize an otherwise intuitive negative-feedback mechanism.

C. Pole interpretation and steady-state regulation

The closed-loop poles are

$$p_1 = -0.25, \quad p_{2,3} = -1.16667 \pm 1.75594\mathrm{j} \quad [\mathrm{s^{-1}}]. \tag{21}$$

The complex pair has $\omega_n = 2.1082\,\mathrm{rad/s}$ and $\zeta = 0.5534$. It governs the early valve motion, while the real pole gives a 4 s tail time constant. Therefore a second-order overshoot formula based only on ζ does not predict the complete speed response; disturbance-transfer zeros and the third pole matter. In particular, speed does not cross nominal in the simulated linear cases, even though throttle position overshoots its final change.

For a constant step d_0, the final-value theorem applied to (17) gives

$$e_\infty = 0, \qquad v_\infty = \frac{d_0}{k}. \tag{22}$$

This holds only when the linear loop is stable, the siphon reference is constant, the pump has negligible slip, and the valve and column have sufficient travel. For $d_0 = \pm 0.3$, the final valve positions are 0.575 and 0.425. The corresponding water-level change is

$$\eta_\infty = -\frac{v_\infty}{K_s \rho g} = \mp 7.645\,\mathrm{mm}. \tag{23}$$

An ordinary fixed drain would make outflow depend on h, replacing ideal integration with a leaky memory and reintroducing permanent speed error. The floating siphon is consequently essential, rather than decorative.

V. Simulation and Results

A. Reproduction and metric definitions

GNU Octave 10.3.0 integrates the linear state model with ode45, relative tolerance 10^{-10}, absolute tolerance 10^{-12}, and output spacing 0.002 s over 80 s. No control toolbox is required. All states start at the nominal equilibrium. Two independent experiments apply process-load steps $T_L/T_{L0} : 1 \to 1.3$ and $1 \to 0.7$ at $t = 0$.

The script also checks the linear trajectory against the matrix-eigenvalue solution. Their maximum state discrepancy is below 2×10^{-12}. The provided CSV contains both speed and valve position for both load directions and both feedback configurations. Fig. 3 uses those results; plots stop at 25 s for readability.

Speed settling time is the first time after which the speed stays within a stated band around its final value.

Since the closed-loop initial and final speeds are both one, the conventional 2% of final speed-change definition is degenerate. We therefore report both a ± 0.01 nominal-speed band and a band equal to 2% of the peak speed excursion. Valve settling uses 2% of its nonzero final opening change. Speed "overshoot" is reported as the peak departure from nominal and separately as crossing beyond nominal.

TABLE II
Linear step-load performance (time in seconds)

Metric	Open loop	Closed +30%	Closed −30%
Final speed	0.700 / 1.300	1.000	1.000
Peak speed departure	30.00%	4.002%	4.002%
Time of speed extremum	Asymptotic	1.016	1.016
Speed crossing nominal	None	None	None
Final valve position	0.500	0.575	0.425
Extreme valve position	0.500	0.58430	0.41570
Valve change overshoot	N/A	12.40%	12.40%
Speed: 1% nominal band	Never	5.548	5.548
Speed: 2% peak band	15.65*	15.630	15.630
Valve: 2% change band	N/A	2.770	2.770

*Open-loop settling is around its shifted final speed; it does not indicate recovery to nominal. Open-loop increase/decrease values are separated by a slash.

Open-loop speed is $1-d_0(1-e^{-t/4})$ and remains offset by 30%. Feedback reduces the maximum excursion by 86.7% and eliminates the ideal steady-state error. The speed-band settling times in the table use different centers for open and closed loop; their similar numerical values do not mean similar regulation quality.

B. Nonlinear and parameter checks

A secondary simulation retains the square-law restriction and the actuator-displacement continuity equation. It solves the monotonic implicit actuator equation at every derivative evaluation; no valve limit is reached. The largest nonlinear-versus-linear speed discrepancy is 0.000504 for added load and 0.000487 for removed load, or at most 0.051 percentage points of nominal speed. Nonlinear extrema are 0.95980 and 1.03983. This checks the local orifice approximation, not friction, steam dynamics, or a real prototype.

A 64-corner sweep independently changes J, B, K_t, K_s, A_t, and τ by $\pm 20\%$, recomputing the physically coupled gains at every corner. Every tested loop remains stable; the worst pole real part is $-0.1871\,\mathrm{s}^{-1}$. This is a finite test, not a proof for every parameter in the continuous uncertainty region.

VI. Practical Realism and Failure Modes

A. Construction and commissioning

A 100 mm bore column provides room for a hollow copper float and a light, rigid siphon. The siphon rises over the column rim and descends to a free outlet; both limbs are carried by the float. Provide approximately $\pm 50\,\mathrm{mm}$ guided travel around $h_0 = 0.40\,\mathrm{m}$. A receiving trough below the lowest outlet position prevents submergence over that travel. The float's buoyancy must support its own weight and the moving pipework with sufficient reserve; mechanical balance can reduce the required immersion.

Use an opposed or double-acting crank pump with leather flap check valves. Flow is cycle-averaged, so the actual reciprocating-flow ripple must be reduced using multiple phased pump strokes or a small air vessel. Added compliance introduces extra dynamics; its time scale should be measured and included before commissioning. The calculations do not certify ripple-free operation.

Before starting, prime the siphon, fill and bleed the pressure line, set the column at h_0, and calibrate its drain to Q_0 by timed volume collection. With the engine held at Ω_0, adjust the spring/counterweight bias until $x_v = 0.5$. A manual startup opening is necessary because the stopped-engine state is outside the chosen linearization. Engage automatic operation near the nominal speed; the design then operates without a human continually moving the throttle.

B. Friction, reference drift, and travel limits

At the final added load, diaphragm travel is only $L(0.075) = 1.5\,\mathrm{mm}$ and the pressure change is $75\,\mathrm{Pa}$, corresponding to $0.15\,\mathrm{N}$ incremental force. Leather friction, valve torque, or a sticking float can therefore dominate the predicted small-signal motion. A broad diaphragm, short low-friction pivots, counterbalancing, and a steam-pressure-balanced valve are necessary. Their effectiveness is an unverified construction assumption.

Pump slip changes the speed-to-flow calibration. A 1% error in reference outflow produces approximately a 1% equilibrium speed error for an otherwise ideal displacement pump, because $D\Omega_\infty = Q_r$. Mineral deposits, a dirty nozzle, water-temperature changes, or a submerged siphon outlet alter that reference. Use filtered water, a removable restriction, an open free jet, and periodic timed-flow checks.

Siphon loss of prime stops the reference drain and can drive the level upward until the engine closes down. A broken pressure line instead unloads the diaphragm and may command excessive steam. An independent mechanical high-speed trip and a weight-closed stop valve should shut off steam on control failure. Such a trip is separate from the proposed continuous governor and has not been dynamically modeled.

At sustained overload or after a large startup error, the valve or column can saturate while inventory continues accumulating. Physical overflow and low-level bypasses can bound the stored volume and ease reset, but they alter the model when active. Their thresholds should lie outside normal travel. The reported experiments remain well inside both the $[0, 1]$ valve range and the proposed column travel.

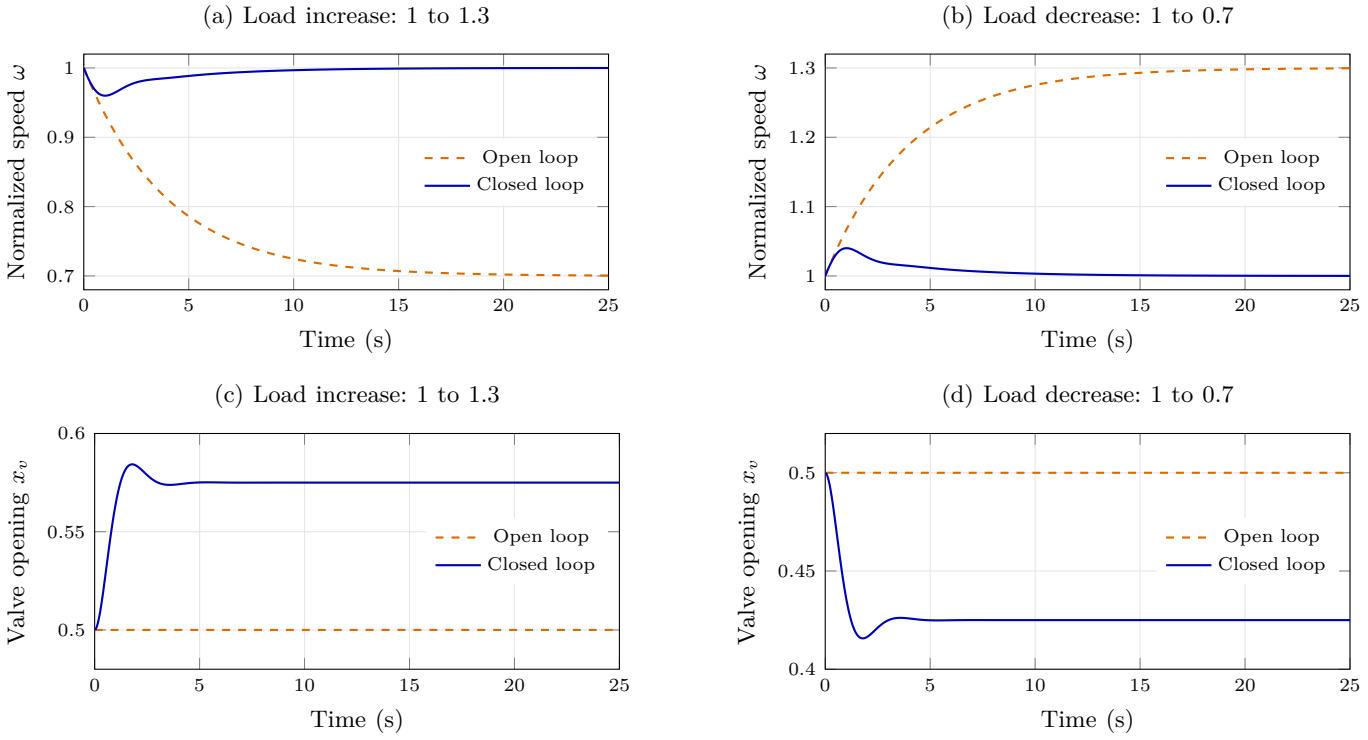


Fig. 3. Required normalized simulations. Dashed orange: fixed-throttle open loop; solid blue: closed loop. Both experiments start at $\omega = 1$, $x_v = 0.5$; process-load changes occur at $t = 0$. The valve position is a fraction of full opening.

VII. Conclusions

The proposed governor senses shaft speed through displaced water flow and stores its corrective action as water inventory. A floating constant-head siphon allows the inventory state to retain a different throttle setting at the same final speed. The derived controller includes diaphragm displacement and predicts stable rejection of the prescribed load steps: 4.00% peak speed excursion and zero ideal steady-state offset. Simplicity lies in using pressure addition and volume conservation rather than a rotating flyweight comparator.

The work establishes a mathematically consistent design proposal, not experimental feasibility or universal originality. The highest-priority next steps are measuring diaphragm friction and valve torque, calibrating the moving siphon, and testing pump ripple and waterline dynamics on a low-power rig. Those measurements would determine whether the predicted advantages survive realistic construction.

Reproducibility: Public repository URL: awaiting team-provided URL. The accompanying source bundle contains the Octave model, parameters, numerical outputs, and standalone LaTeX source. A public URL must replace this statement before submission.

References

[1] P. Agarwal, "ME319: Control Systems, Autumn 2026, Assignment 1: Reinventing the Steam Engine Governor," IIT Bombay, pp. 1–5, 2026. Assignment supplied by the team.

[2] E. B. Chase and F. N. Payne, compilers, Selected Techniques in Water Resources Investigations, 1966–67, U.S. Geological Survey Water-Supply Paper 1892, p. 22, Fig. 5, 1968. [Online]. Available: https://pubs.usgs.gov/wsp/1892/report.pdf. Accessed: Oct. 15, 2026.

[3] Woodward, EGB-10PLS Hydraulic Actuator: Installation and Operation Manual, Manual 82558, Rev. A, Sec. "Mechanical Governor," pp. 12–13. [Online]. Available: https://www.woodward.com/products/publication-asset/82558/. Accessed: Oct. 15, 2026.

Mandatory AI Appendix: Tools, Prompts, and Critical Decisions

Scope and authorship transparency. This appendix records AI assistance in preparing this document. The named team members were supplied by the user. No individual contribution percentages, student experiments, prototype measurements, or independent student verification have been inferred. All technical statements and calculations remain subject to the team's review before submission.

A. Models and software actually used

OpenAI Codex, a GPT-6-based assistant, developed the concept, derivations, source code, prose, and schematic. Web search was used for a limited comparison with existing hydraulic governors and to check the floating-siphon principle against USGS material. GNU Octave 10.3.0 executed the simulations, computed poles and response metrics, checked a nonlinear orifice model, and evaluated 64 parameter corners. Python with NumPy independently checked pole values and prepared plotting coordinates from Octave output; pypdf extracted the supplied assignment, and pypdfium2 rendered PDF pages for visual inspection. The Codex built-in LaTeX compiler checked the standalone source; Tectonic exported the PDF. TikZ and PGFPlots generated vector diagrams and curves. No symbolic-computation package was used.

B. Prompts and how they were used

The user requested: "members of the team are Dhanvin A.S 24B2261, Varad 24B2179, Rahul Sunda 24B2147" and "Read all the instructions and generate the PDF in latex and give me." These supplied the author information and requested deliverable. The following are summaries of the assistant's internal work directions, not additional prompts claimed to have been written by the students: (1) extract all five assignment pages and identify the required sections, normalization, plots, and submission conditions; (2) propose a period-plausible hydraulic feedback mechanism, identify similarities to known governors, and avoid claiming unverified novelty; (3) derive flow continuity, pressure and actuator equations, retaining the water displaced by actuator motion; (4) simulate both prescribed step-load cases in Octave, independently check the linear solution, and compare against a nonlinear orifice model; (5) typeset an IEEE paper with original vector diagrams, quantified limitations, and this appendix, then compile and inspect it.

C. AI outputs accepted, modified, and rejected

Disposition	Output and engineering reason
Accepted	The pump–restriction–inventory-column concept with a floating siphon: pump flow senses speed, the restriction creates a prompt pressure response, and stored volume can preserve the throttle correction after speed recovery. The claims are provisional design findings.
Modified	The initial signal-only actuator approximation was replaced by $Q_p = q - \alpha \dot{x}_v$. The resulting coefficients are $a = 0.45$ s and $c = 1.05$, rather than 0.25 s and 1; displaced fluid materially changes the controller dynamics.
Modified	A preliminary lower-damping engine estimate was replaced by an explicitly stated effective drag estimate, $B = 10$ N m s, so the fixed-throttle comparison remains physically interpretable for both prescribed steps. This is a design assumption, not an experimentally fitted value.
Rejected	A fixed bottom drain was rejected because its head-dependent outflow erodes integral memory. A long viscous capillary was rejected in favor of a thin restriction to reduce viscosity sensitivity and neglected line inertance.
Rejected	Unqualified claims of historical originality, built hardware, measured performance, and standard second-order overshoot predictions were excluded. No fabricated public repository URL or team contribution breakdown was inserted.

D. Validation, interpretation, and remaining human decisions

The Routh calculation and reported poles agree; the Octave ODE solution agrees with an exact eigenvalue solution to below 2×10^{-12} in the simulated states. The nonlinear check differs by at most 0.051 percentage points in speed, but tests only the selected loss law. Speed settling bands are defined explicitly because initial and final regulated speeds coincide. The 64-corner check establishes stability only at the sampled corners. Engineering estimates for damping, discharge coefficients, diaphragm friction, siphon tracking, pump slip, and pressure-balanced valve torque require experimental validation. Limited web comparison does not establish worldwide novelty. The team must review the derivation and reproducibility bundle, supply a public GitHub URL, and independently complete the confidential peer evaluations; no such external submissions have been performed by the assistant.